# Tumor necrosis factor-α inhibitor-induced psoriasis: Systematic review of clinical features, histopathological findings, and management experience

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***Background:*** Tumor necrosis factor-α (TNF-α) inhibitors have been reported to induce new-onset psoriasis.

***Objective:*** To better define the demographic, clinical features, and treatment approach of TNF-α inhibitor-induced psoriasis.

***Methods:*** Systematic review of published cases of TNF-α inhibitor-induced psoriasis.

***Results:*** We identified 88 articles with 216 cases of new-onset TNF-α inhibitor-induced psoriasis. The mean age at psoriasis onset was 38.5 years. The most common underlying diseases were Crohn disease (40.7%) and rheumatoid arthritis (37.0%). Patients underwent TNF-α therapy for an average of 14.0 months before psoriasis onset with 69.9% of patients experiencing onset within the first year. The majority of patients received skin-directed therapy, though patients who discontinued TNF therapy had the greatest resolution of symptoms (47.7%) compared with those who switched to a different TNF agent (36.7%) or continued therapy (32.9%).

***Limitations:*** Retrospective review that relies on case reports and series.

***Conclusion:*** While TNF-α inhibitor cessation may result in resolution of induced psoriasis, lesions may persist. Decisions regarding treatment should be weighed against the treatability of TNF-α inhibitor-induced psoriasis, the severity of the background rheumatologic or gastrointestinal disease, and possible loss of efficacy with cessation followed by retreatment. Skin-directed therapy is a reasonable initial strategy except in severe cases. (J Am Acad Dermatol 2017;76:334-41.)

***Key words:*** adalimumab; adverse event; certolizumab; etanercept; golimumab; infliximab; medication side effect; psoriasis; tumor necrosis factor-α-induced psoriasis; tumor necrosis factor-α inhibitor.

The proinflammatory cytokine tumor necrosis factor-α (TNF-α) has been implicated in the pathogenesis of multiple inflammatory and autoimmune conditions such as Crohn disease (CD), ulcerative colitis (UC), rheumatoid arthritis (RA), ankylosing spondylitis (AS), and psoriasis. The development of TNF-α inhibitors has dramatically improved therapeutic options for patients with these conditions. However, there are many reports in the literature of TNF-α inhibitors paradoxically inducing new onset psoriasis or worsening pre-existing quiescent psoriatic disease. A study analyzing the United States Food and Drug Administration Adverse Event Reporting System from 2004-2011 found that TNF-α inhibitors used in the treatment of Crohn Disease were associated with an increased risk of

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psoriasiform adverse events compared to other medications.[1] There is currently no standardized approach for managing TNF-α inhibitor-induced psoriasis, and there is little clarity on whether TNF-α inhibitor therapy should be withdrawn.

The objective of this systematic review of reported cases of TNF-α inhibitor-induced psoriasis was to better define the demographic, clinical, and histological features of TNF-α inhibitor-induced psoriasis and determine the optimal treatment approach.

## CAPSULE SUMMARY

- The paradoxical development of psoriasis is an unintended consequence of TNF-α inhibition.
- Discontinuing TNF-α therapy resulted in psoriasis resolution (47.7%) more often than switching (36.7%) or continuing (32.9%) TNF-α therapy.
- TNF-α inhibitor-induced psoriasis is often successfully managed with skin-directed therapies and in many cases does not require cessation of TNF-α—inhibitor treatment.

## METHODS

We conducted an electronic literature search to identify studies, case reports, and case series that documented new-onset psoriasiform lesions in patients being treated with a TNF-α inhibitor (Fig 1).

### Information sources and search strategy

With the assistance of a research librarian, we identified literature on PubMed on June 18, 2013 using the terms "anti—TNF-alpha", "TNF-alpha inhibitor", "TNF-alpha inhibitors", "TNF-alpha antagonist", "TNF-alpha antagonists", "anti-tumor necrosis factor", "tumor necrosis factor inhibitor", "tumor necrosis factor inhibitors", "tumor necrosis factor antagonist", "tumor necrosis factor antagonists", "infliximab", "adalimumab", "etanercept", "golimumab", "certolizumab", "psoriasis", "psoriasiform"(medical subject headings [MeSH]), "palmoplantar"(MeSH), "pustul*", "induce*", "cause*", "complicat*", "due", "flar*", "exacerbat*", "paradox*", "induct*", "induce*", "advers*", "onset", "associat*", "risk", "non-psoriatic", "psoriasis/chemically induced"(MeSH), "Tumor necrosis factor-alpha/adverse effects"(MeSH). We restricted the search results to English only records. We conducted similar computerized searches using SCOPUS and the Cochrane Database. The last search was conducted on July 1, 2014. Additionally, we reviewed citations within the identified articles and relevant reviews to locate published articles missed by database searches. We used EndNote to identify and remove duplicates.

### Study selection

Two investigators (MH and MW) assessed study eligibility using title and abstract for initial screening. Two additional investigators (EW and GB) further evaluated the eligibility of the studies by reviewing the full-text publication. Any study that reported individual data of patients with new-onset psoriasis after the initiation of a TNF-α inhibitor was eligible for inclusion. We excluded records that omitted specific data regarding individual patients, as well as cases in which the patient had a prior history of psoriasis.

### Data collection and extraction

Two reviewers (EW and GB) independently extracted individual patient data from each record. We constructed a data collection spreadsheet and extracted the following data items from each record: patient demographics (including age, gender, ethnicity), age at onset of psoriasiform eruption, family history of psoriasis, prior TNF-α inhibitor medication history, disease treatment, TNF-α inhibitor therapy resulting in psoriasiform eruption, time elapsed before onset of psoriasiform eruption, morphology of psoriasiform lesions, body areas involved, biopsy results, recurrence of disease after switching to a different TNF-α inhibitor, concomitant immunomodulator use, concomitant systemic steroid use, management, and response.

## RESULTS

### Clinical and histopathologic characteristics

Of the 190 full-text articles assessed for eligibility, 88 articles met inclusion criteria for the final analysis. We extracted a total of 216 cases of new-onset TNF-α inhibitor-induced psoriasis from these publications. Demographic features of patients with TNF-α inhibitor-induced psoriasis are summarized in Table I.

Women comprised 72.2% of the cases, with a female predominance in RA and CD (89.9% and 63.2%, respectively). Ages of psoriasis onset ranged from 7 years to 83 years; the mean age of psoriasis onset was 38.5 years. The majority of patients received TNF-α inhibitor therapy for CD (40.7%), followed by RA (37.0%) and AS (13.9%). Patients had a positive family history of psoriasis in 11.8% (N = 19) of disclosed cases (N = 161), and no patients had a personal history of psoriasis or psoriatic arthritis because those cases were excluded from analysis. There were no identified cases with known preexisting psoriatic arthritis before developing psoriasis on TNF-α inhibitor therapy.

*Abbreviations used:*

| | |
|---|---|
| AS: | ankylosing spondylitis |
| AZA: | azathioprine |
| CD: | Crohn disease |
| IFN-$\alpha$: | interferon $\alpha$ |
| IL: | interleukin |
| IL-23R: | interleukin-23 receptor |
| MeSH: | medical subject headings |
| MTX: | methotrexate |
| RA: | rheumatoid arthritis |
| $T_H$: | T helper |
| TNF-$\alpha$: | tumor necrosis factor-$\alpha$ |
| UC: | ulcerative colitis |

The psoriasis presentations were variable, and 26.9% of patients had more than one reported morphological type. The two most commonly observed clinical presentations included plaque (44.8%) and palmoplantar pustular (36.3%) psoriasis. Other presentations included ill-defined psoriasiform dermatitis (19.9%), severe scalp involvement associated with alopecia (7.5%), and generalized pustular psoriasis (10.9%). The anatomical sites commonly involved included soles (45.8%), extremities (45.4%), palms (44.9%), scalp (36.1%), and trunk (32.4%).

Of the 102 cases confirmed with biopsy, 54.9% revealed plaque psoriasis and 33.3% were consistent with pustular psoriasis with the remainder interpreted as psoriasiform dermatitis. An eosinophil-rich infiltrate was noted in 3 of the plaque psoriasis biopsies and 3 of the pustular psoriasis biopsies.

### TNF-$\alpha$ inhibitor treatment characteristics

The majority of cases were associated with infliximab (62.5%), followed by adalimumab (21.8%) and etanercept (14.4%). Only 2 cases involved certolizumab and 1 case with golimumab. The most commonly prescribed agent was infliximab (62.5%), with the majority of patients with CD (82.8%) and AS (73.3%) receiving infliximab compared to other agents. Patients underwent TNF-$\alpha$ inhibitor therapy for an average of 14.0 months before onset (infliximab 13.6 months; adalimumab 14.4 months; etanercept 16.2 months; certolizumab 5.0 months; golimumab 4.0 months; overall range 1-120 months). Of the 206 cases reporting the time of psoriasis onset, 69.9% (144/206) of patients experienced the onset of psoriasis in the first year of treatment; 31.2% of cases occurred in the second year (infliximab 17.4%; adalimumab 9.8%; etanercept 13.3%). In total, 19 patients reported prior TNF-$\alpha$ inhibitor treatment, 3 of which had previously experienced psoriasiform eruption and resolution with withdrawal. The most common concomitant immunomodulators at the time of psoriasis onset included methotrexate (MTX; 33.7%), azathioprine (AZA; 18.9%), and leflunomide (13.7%). Seventeen cases reported administration of concomitant systemic steroids.

### Management and outcomes

Therapeutic management and outcomes of TNF-$\alpha$ inhibitor-induced psoriasis are summarized in Table II. Resolution of psoriasis (or no evidence of recurrence at time of follow-up) was reported in patients who discontinued TNF therapy (47.7%), switched to a different TNF agent (36.7%), and continued the same TNF therapy (32.9%). Improvement but incomplete resolution of psoriasis was reported in patients who continued the same TNF therapy (57.3%), discontinued TNF therapy (46.2%), and switched to a different TNF agent (18.4%). Regardless of the TNF-$\alpha$ inhibitor treatment decision, the majority of patients received skin-directed therapy with one or more agents including topical steroids (76.5%), vitamin D analogues (17.6%), MTX (17.2%), phototherapy (8.3%), cyclosporine (5.4%), acitretin (2.0%), and coal tar (1.0%).

Management strategies and resulting outcomes were analyzed for patients with more severe presentations, including alopecia and/or generalized pustular psoriasis (Table III). Half of the patients with alopecia had resolution of symptoms regardless of continuing or discontinuing TNF therapy (resolution in 2/4 who continued vs 4/8 who discontinued therapy). Patients with generalized pustular psoriasis who continued therapy experienced resolution in 33.3% of cases (2/6). Of patients with generalized pustular psoriasis who discontinued therapy, 63.6% had improvement of symptoms and 27.3% had resolution of psoriasis at follow-up.

## DISCUSSION

Psoriasiform lesions are a well-documented complication of TNF-$\alpha$ inhibitor therapy that can occur at any time during the treatment course.[2] However, in our analysis, we found the majority (69.9%) occur within the first year of treatment. Furthermore, our review of published photographs suggests that many patients had generalized involvement with more than one morphological type. The typical presentations include plaque or palmoplantar pustular psoriasis, though various clinical morphologies have been reported. Severe scalp psoriasis associated with alopecia is a less common presentation observed in de novo psoriasis and the onset in a patient undergoing TNF-$\alpha$ inhibitor therapy should prompt consideration of TNF-$\alpha$ inhibitor-induced psoriasis.

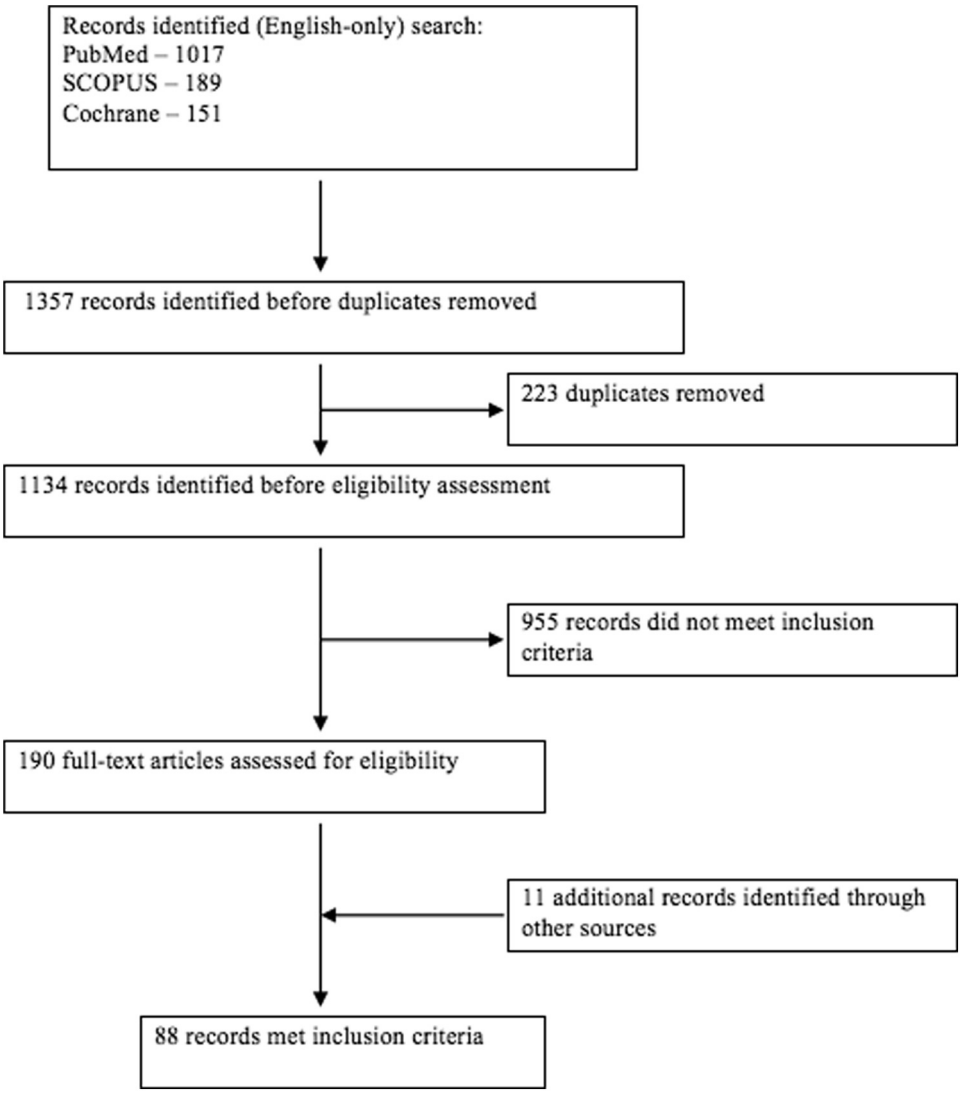


**Fig 1.** The electronic search, selection, and exclusion of studies in the systematic review.

The majority of the TNF-$\alpha$ inhibitor-induced psoriasis biopsy samples were histologically indistinguishable from de novo psoriasis. However, in a study investigating histopathologic changes in TNF-$\alpha$ inhibitor-induced psoriasiform clinical lesions, the authors concluded that the reactions comprise a spectrum ranging from that closely mimicking psoriasis, a sterile pustular folliculitis, or a lichenoid pattern.[3] The presence of eosinophils may also be suggestive of TNF-$\alpha$ inhibitor-induced psoriasiform dermatitis.[3] A biopsy may be helpful in patients with new-onset psoriasis while on TNF-$\alpha$ inhibitors to evaluate for these specific features. Biopsying may also be helpful in distinguishing TNF-$\alpha$ inhibitor-induced psoriasis from lichenoid drug reaction, as both can present with clinically psoriasiform lesions. There are fewer reported TNF-$\alpha$ inhibitor-induced lichen planus cases in the literature; however, the prognosis is comparable to TNF-$\alpha$ inhibitor-induced psoriasis such that most patients completely resolved following agent continuation or withdrawal with concomitant skin-directed treatment.[4]

In this study we found that more patients had resolution of symptoms after discontinuing TNF therapy (47.7%) compared to those who continued therapy (32.9%). While the majority of patients switching to a different TNF-$\alpha$ inhibitor had either resolution (36.7%) or improvement (18.4%) in psoriasis, 44.9% experienced no improvement or recurrence, suggesting a need for additional treatment options in these patients.

The management approach may also vary depending on the clinical presentation. In patients with more severe presentations, such as generalized pustular psoriasis, only 27.3% of patients who discontinued therapy had resolution compared to 47.7% of

**Table I.** Summary of demographic features of patients with TNF-$\alpha$ inhibitor-induced psoriasis

| Demographic characteristics | Patients (N = 216) |
|---|---|
| Gender, N (%) | |
| Female | 156 (72.2) |
| Male | 60 (27.8) |
| Average age, years (range) | 38.5 (7-83) |
| Family history of psoriasis, N (% of 161 reported) | 19 (11.8) |
| Primary disease, N (%) | |
| Crohn disease | 88 (40.7) |
| Rheumatoid arthritis | 80 (37.0) |
| Ankylosing spondylitis | 30 (13.9) |
| Ulcerative colitis | 6 (2.8) |
| Juvenile idiopathic arthritis | 5 (2.3) |
| Behcet disease | 3 (1.4) |
| Spondyloarthropathy | 5 (2.3) |
| Synovitis, acne, pustulosis, hyperostosis, osteitis (SAPHO) | 2 (1.0) |
| Bilateral idiopathic panuveitis | 1 (0.5) |
| Inflammatory arthritis | 1 (0.5) |
| TNF-alpha inhibitor resulting in psoriasis eruption, N (%) | |
| Infliximab | 135 (62.5) |
| Adalimumab | 47 (21.8) |
| Etanercept | 31 (14.4) |
| Certolizumab | 2 (1.0) |
| Golimumab | 1 (0.5) |
| Concomitant immunomodulators, N (% of 95 reported) | |
| Methotrexate | 32 (33.7) |
| Azathioprine | 18 (18.9) |
| Leflunomide | 13 (13.7) |
| Sulfasalazine | 7 (7.4) |
| Hydroxychloroquine | 2 (2.1) |
| 6-Mercaptopurine | 2 (2.1) |
| Mesalazine | 2 (2.1) |
| Cyclosporine | 1 (1.0) |
| Tacrolimus | 1 (1.0) |
| Concomitant systemic steroids, N | 17 |
| Clinical latency to eruption, months (range in months) | |
| Infliximab | 13.6 (1-120) |
| Adalimumab | 14.4 (1-62) |
| Etanercept | 16.2 (2-72) |
| Certolizumab | 5.0 (4-6) |
| Golimumab | 4.0 (4; N = 1) |
| Morphology of skin lesions, N (%) | |
| Plaque | 90 (44.8) |
| Palmoplantar pustular psoriasis | 73 (36.3) |
| Psoriasiform | 40 (19.9) |
| Severe alopecia | 15 (7.5) |
| Generalized pustular | 22 (10.9) |
| Guttate | 16 (8.0) |
| Inverse | 7 (3.5) |
| Follicular | 2 (1.0) |
| More than one form | 54 (26.9) |
| Body areas involved, N (%) | |
| Soles | 99 (45.8) |
| Extremities | 98 (45.4) |
| Palms | 97 (44.9) |
| Scalp | 78 (36.1) |
| Trunk | 70 (32.4) |
| Face | 19 (8.8) |
| Axillae | 15 (6.9) |
| Groin | 14 (6.5) |
| Not specified | 20 (9.3) |
| Histology of lesions, N (% of 102 reported) | |
| Classic | 56 (54.9) |
| Pustular psoriasis | 34 (33.3) |
| Psoriasiform dermatitis | 12 (11.8) |
| Psoriatic treatment, N (% of 204 reported) | |
| Topical steroids | 156 (76.5) |
| Vitamin D analogue | 36 (17.6) |
| Methotrexate | 35 (17.2) |
| Systemic steroids | 19 (9.3) |
| Phototherapy | 17 (8.3) |
| Cyclosporine | 11 (5.4) |
| Acitretin | 4 (2.0) |
| Coal tar | 2 (1.0) |
| Unknown | 12 (5.9) |

*TNF*, Tumor necrosis factor.

all patients who discontinued therapy analyzed in the systematic review. However, 63.6% of patients with generalized pustular psoriasis showed improvement, but no resolution, after discontinuing therapy. Therefore, such patients may need counseling on realistic expectations on the outcome and the probability that they may experience improvement rather than complete resolution of symptoms after discontinuing therapy. In contrast, half of the patients with alopecia had resolution of symptoms despite continuing or discontinuing therapy, albeit a larger quantity of patients were managed with discontinuation of therapy. These results suggest that patients may have different outcomes based on clinical subtype of psoriasis. While the number of patients with alopecia or generalized pustular psoriasis who switched to a different TNF-$\alpha$ agent was small, all experienced either resolution or improvement of symptoms, which suggests that this may be a reasonable therapeutic approach in a subset of patients.

The prevalence of new-onset psoriasis in patients treated with TNF-$\alpha$ inhibitors is unknown. The underlying disease itself may play a predisposing role, as patients with chronic rheumatologic and gastrointestinal inflammatory diseases have a higher incidence of psoriasis.[5,6] Immune dysregulation related to immunosuppression may additionally contribute; the prevalence of TNF-$\alpha$ inhibitor-induced psoriasis

**Table II.** Summary of therapeutic management and outcomes of patients with TNF-$\alpha$ inhibitor-induced psoriasis

| | Total | Infliximab | Adalimumab | Etanercept | Golimumab | Certolizumab |
|---|---|---|---|---|---|---|
| Continued on TNF therapy (N) | 82 | 55 | 14 | 12 | 1 | - |
| Resolved, N (%) | 27 (32.9) | 22 (40.0) | 1 (7.1) | 4 (33.3) | - | - |
| Improved, N (%) | 47 (57.3) | 30 (54.5) | 12 (85.7) | 4 (33.3) | 1 (100.0) | - |
| No improvement, N (%) | 8 (9.8) | 3 (5.5) | 1 (7.1) | 4 (33.3) | - | - |
| Discontinued off TNF therapy (N) | 65 | 32 | 19 | 12 | - | 2 |
| Resolved, N (%) | 31 (47.7) | 17 (53.1) | 10 (52.6) | 3 (25.0) | - | 1 (50.0) |
| Improved, N (%) | 30 (46.2) | 12 (37.5) | 8 (42.1) | 9 (75.0) | - | 1 (50.0) |
| No improvement, N (%) | 4 (6.2) | 3 (9.4) | 1 (5.3) | - | - | - |
| Switched to different TNF agent (N) | 49 | 9 | 19 | 18 | - | 3 |
| Resolved/no recurrence at follow-up, N (%) | 18 (36.7) | 2 (22.2) | 8 (42.1) | 7 (38.9) | - | 1 (33.3) |
| Improved, N (%) | 9 (18.4) | 2 (22.2) | 2 (10.5) | 4 (22.2) | - | 1 (33.3) |
| No improvement, N (%) | 22 (44.9) | 5 (55.6) | 9 (47.4) | 7 (38.9) | - | 1 (33.3) |

*TNF*, Tumor necrosis factor.

**Table III.** Summary of outcomes with severe presentations including alopecia and/or generalized pustular psoriasis

| Clinical subtype* | Clinical outcome | Continued TNF therapy, n/N (%) | Discontinued TNF therapy, n/N (%) | Switched to different TNF agent, n/N (%) |
|---|---|---|---|---|
| Alopecia | Resolved | 2/4 (50.0%) | 4/8 (50.0%) | |
| | Improved | 1/4 (25.0%) | 2/8 (25.0%) | 1/1 (100.0%) |
| | No improvement | 1/4 (25.0%) | 2/8 (25.0%) | |
| | Unknown (N = 2) | | | |
| Generalized pustular psoriasis | Resolved | 2/6 (33.3%) | 3/11 (27.3%) | 2/4 (50.0%) |
| | Improved | 2/6 (33.3%) | 7/11 (63.6%) | 2/4 (50.0%) |
| | No improvement | 2/6 (33.3%) | 1/11 (9.1%) | |
| | Unknown (N = 1) | | | |

*TNF*, Tumor necrosis factor.
*Note that 3 patients had both alopecia and generalized pustular psoriasis.

in patients with RA and spondyloarthropathy (2.3%-5%)[7-9] is similar to that in HIV disease (1.3%-1.5%).[10] However, the temporal relationship between TNF-$\alpha$ inhibitor and psoriasis onset (most often in the first year after exposure) supports a causal association. Overall, the data suggest that TNF-$\alpha$ inhibitor-induced psoriasis is a class effect, as this adverse reaction has been observed with all TNF-$\alpha$ inhibitors and with all diseases for which TNF-$\alpha$ inhibitors are indicated.[9,11]

While the exact mechanism of TNF-$\alpha$ inhibitor-induced psoriasis remains elusive, emerging evidence suggests that the interleukin (IL)-23/T-helper ($T_H$)17 axis plays an important role in the pathogenesis.[12] IL-23 is a proinflammatory cytokine that drives downstream $T_H1$ and $T_H17$ effector responses, which have been implicated in the pathogenesis of chronic inflammatory diseases.[13] Genome-wide association studies have associated specific polymorphisms in the IL-23 receptor (IL-23R) gene with increased susceptibility to both CD and psoriasis.[14,15] In a study evaluating TNF-$\alpha$ inhibitor-induced psoriasis in pediatric patients with CD, those who developed psoriasis were more often homozygous for three specific IL-23R polymorphisms compared to disease-matched controls who did not develop psoriasis following TNF-$\alpha$ inhibitor treatment.[16] Similarly, adult patients with severe TNF-$\alpha$ inhibitor-induced psoriasis and psoriatic alopecia were homozygous wild type carriers for a rare IL-23R variant.[17] Consistent with molecular studies, IL-12/23 antagonism with ustekinumab has been an effective treatment in several CD patients who developed TNF-$\alpha$ inhibitor-induced psoriasis.[17,18] In addition, IL-23 regulates $T_H17$ cells, which secrete IL-17, a cytokine known to play a critical role in psoriasis. IL-17 serum and lesional levels in those with plaque and pustular psoriasis are higher as compared to controls (those without psoriasis).[19] Furthermore, IL-17—expressing T cell infiltrates have been identified in TNF-$\alpha$ inhibitor-induced psoriatic lesions.[17,20] In a study by Tillack et al, patients with

CD who required transition to ustekinumab expressed high levels of infiltrating IL-17A$^+$ cells.[17] While ustekinumab is not currently approved by the Food and Drug Administration for treatment of CD, clinical trials have shown promising results.[21-23] This data suggests that IL-23 and/or IL-17 antagonism may benefit a patient subset with severe TNF-$\alpha$ inhibitor-induced psoriasis.

Disequilibrium in proinflammatory cytokine interferon $\alpha$ (IFN-$\alpha$) levels in the setting of TNF-$\alpha$ suppression is also thought to contribute to psoriatic lesion development.[24-26] TNF-$\alpha$ normally attenuates IFN-$\alpha$ levels, therefore TNF-$\alpha$ inhibition results in elevated IFN-$\alpha$. IFN-$\alpha$ stimulates $T_H1$ lymphocytes, which play a role in the pathogenesis of psoriasis.[27] Furthermore, patients with hepatitis who developed psoriasis during IFN-$\alpha$ treatment demonstrated regression of psoriatic lesions following IFN agent withdrawal.[28,29] Moreover, increased IFN-$\alpha$ expression has been demonstrated in lesional dermal vasculature in patients with TNF-$\alpha$ inhibitor-induced psoriasis.[24] Therefore, IFN-$\alpha$ dysregulation may contribute to the pathogenesis of TNF-$\alpha$ inhibitor-induced psoriasis.

The primary limitation of this review was variations in data presented in the reviewed publications. Not all authors performed biopsies or disclosed treatment approach or outcome. In addition, the follow-up interval varied among papers and may not truly represent the eventual clinical outcome. Furthermore, the long latency period between TNF-$\alpha$ inhibitor initiation and psoriasis onset does not exclude the possibility of de novo psoriasis onset occurring independently of TNF-$\alpha$ inhibitor therapy. The data is limited by the retrospective nature and reliance on case reports and small case series. Patients with inflammatory bowel or rheumatologic disease requiring TNF-$\alpha$ inhibitor therapy are at risk for progression of disease if effective medications are discontinued after development of psoriasis. Identifying genotypic or phenotypic features of disease or risk factors for development of TNF-$\alpha$ inhibitor-induced psoriasis may allow for the risk stratification of patients. For example, with inflammatory bowel disease, the majority of patients who reported the development of psoriasis had CD rather than UC, which may be due to the stronger association of CD and psoriasis with *IL-23R* and *IL-12B* gene variants than UC is with these genes, as previously mentioned. In addition, the majority of affected patients are female. Beyond that, with case reports and case series, only the features of affected patients are presented, thus it is not clear whether there is a difference in disease location, phenotype, or activity between patients who seek treatment for psoriasis and others, who may not seek treatment for their symptoms.

## Conclusion

The paradoxical development of psoriasis can be an unintended consequence of TNF-$\alpha$ antagonism. TNF-$\alpha$ inhibitor cessation may result in resolution of induced psoriasis in nearly half the cases, but there is still a substantial proportion of patients for whom lesions may persist. Decisions regarding interruption of anti—TNF-$\alpha$ therapy should be carefully considered in light of the possibility of psoriasis persistence and possible loss of efficacy for rheumatologic or gastrointestinal disease with cessation, should rechallenge become necessary. Thus, an initial approach of treating through with typical skin-directed therapies is reasonable except in the most severe cases. Further research is needed to clarify optimal strategies and clinical outcomes based on different clinical subtypes. While much remains to be learned about the mechanism of TNF-$\alpha$—induced psoriasis, emerging data on specific IL-23R genetic polymorphisms have revealed potential therapeutic targets for patients with severe disease.

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